JUNIA Grande école d'ingénieurs
HEI·ISEN·ISA

Septembre -Décembre 2025

ISEN
Module
Numérique Responsable

Today



End of life

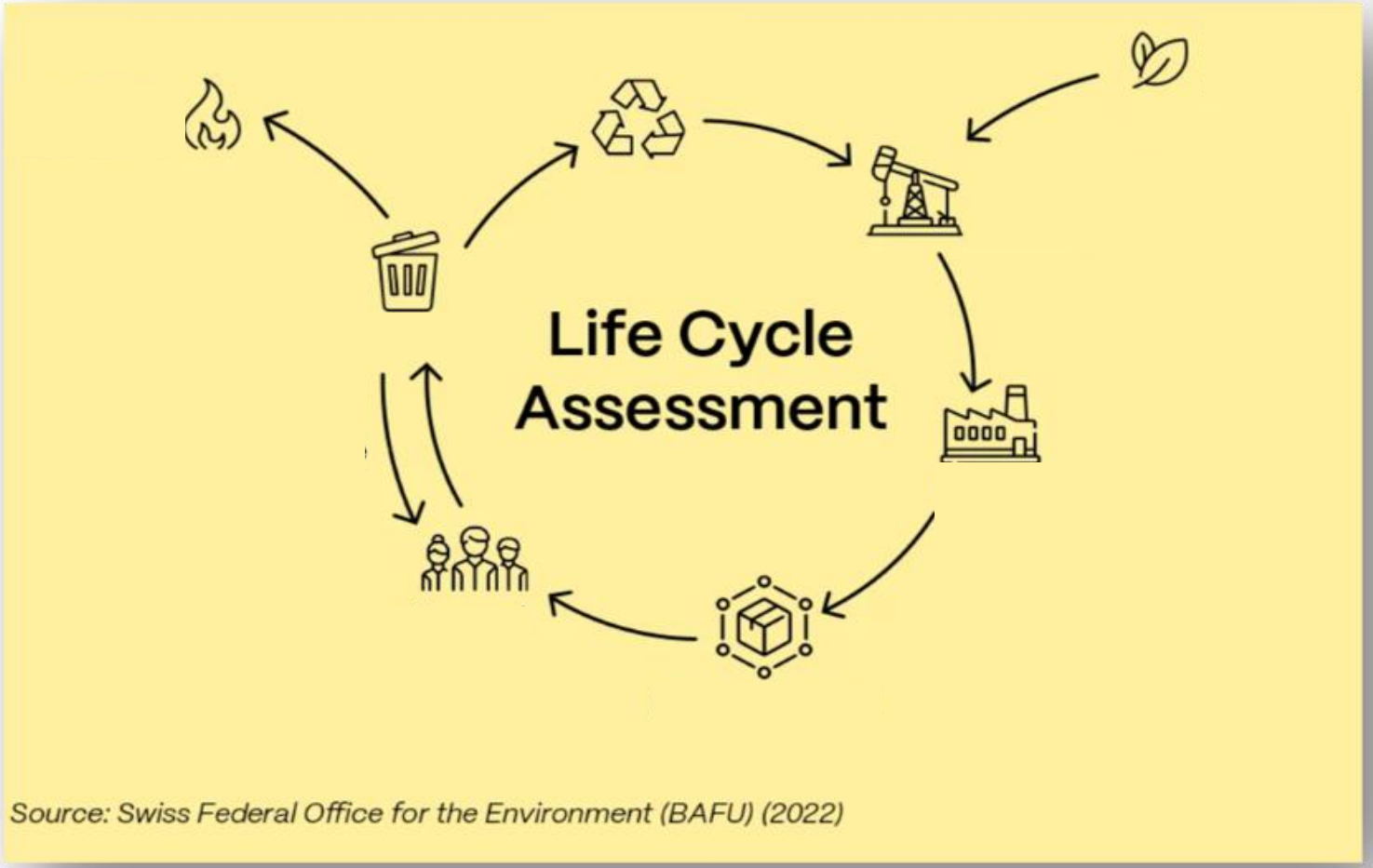
Regulation

Inclusive digital

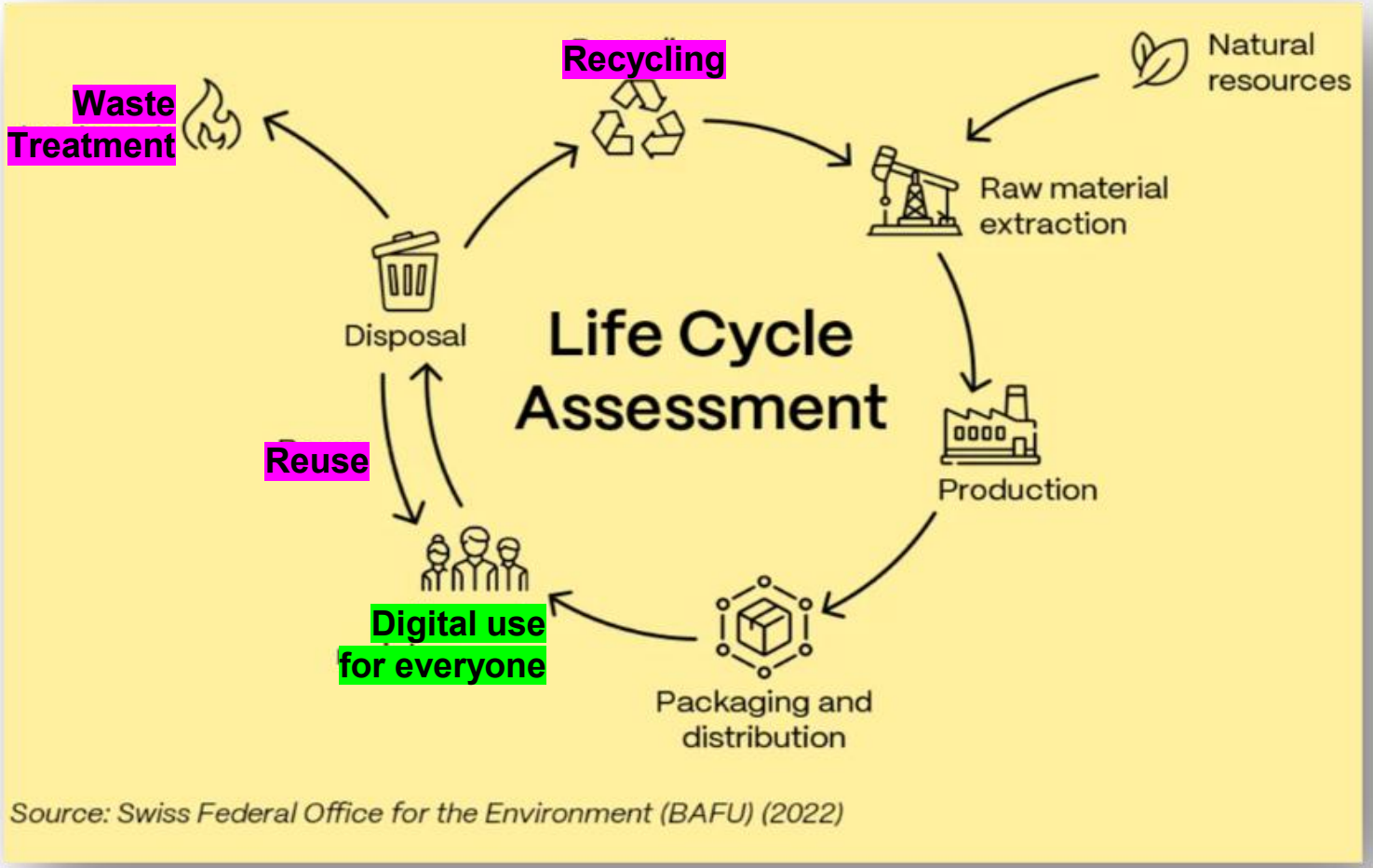
Best practices



Do you remember the lifecycle steps ?



Source: Swiss Federal Office for the Environment (BAFU) (2022)



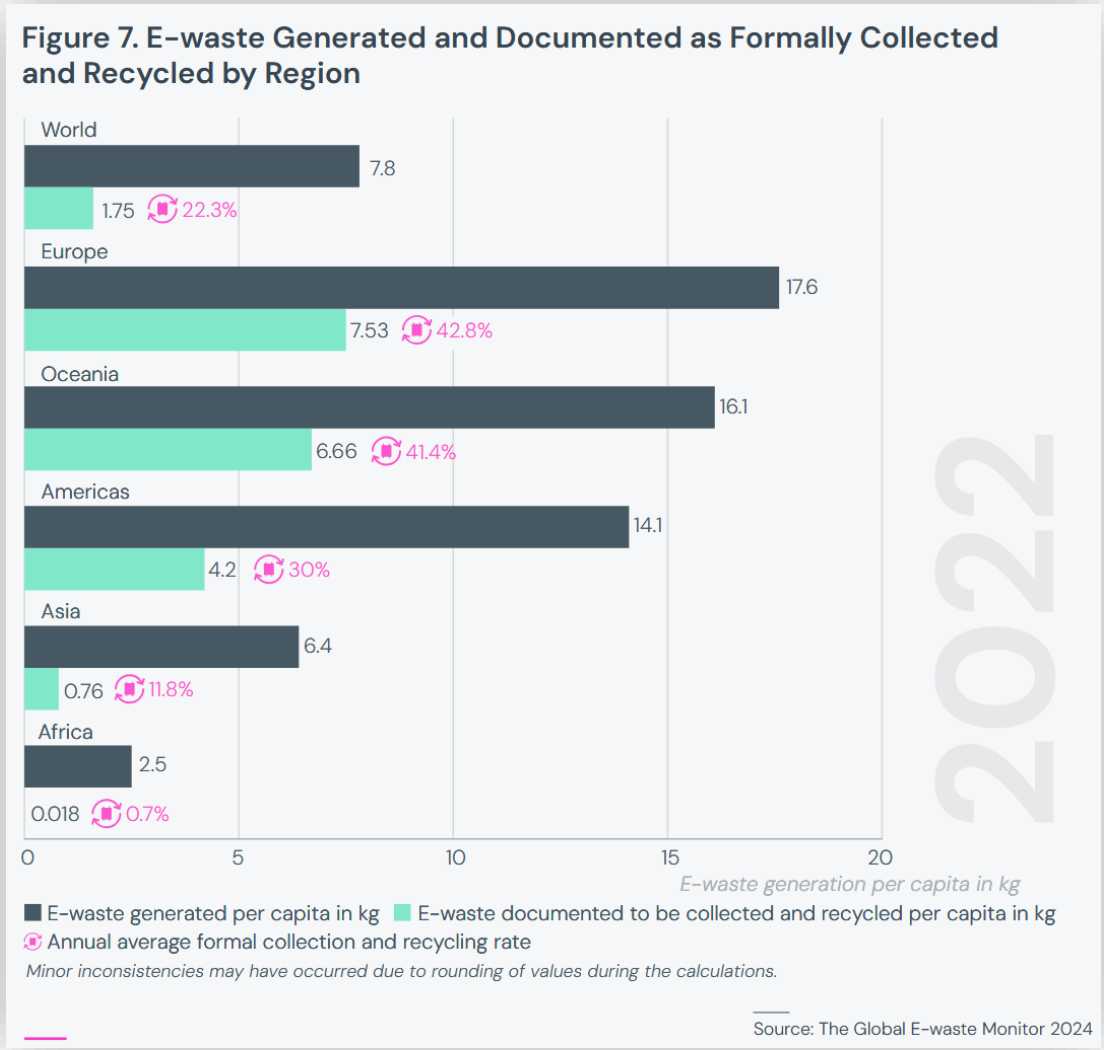
1/ Repare, Recycle, Waste

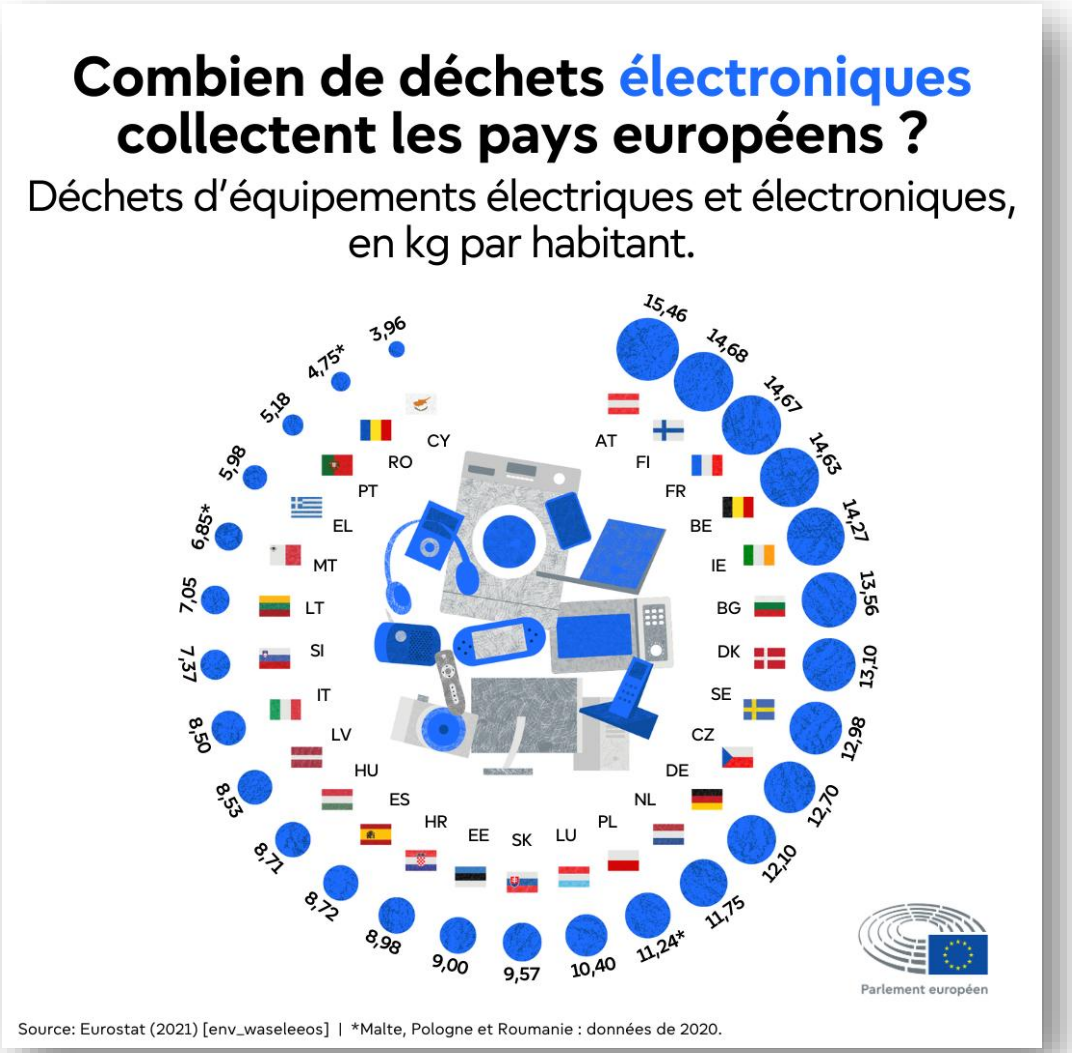
2/ Inclusive digital : Accessibility

REPARE - REUSE - RECYCLE – WASTE



Few figures about end of life





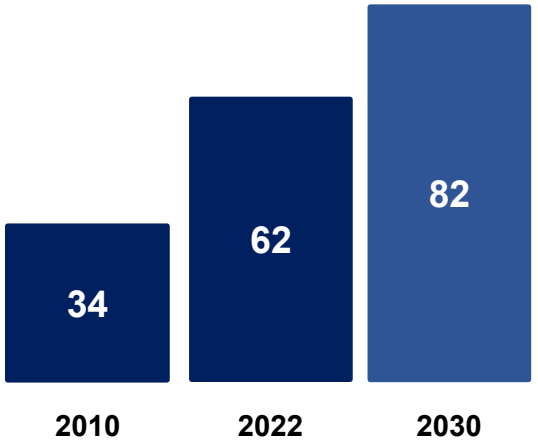
Sources : https://ewastemonitor.info/the-global-e-waste-monitor-2024/, Parlement Européen

E-waste weight...

World "e-waste"

Millions tons





7,8 kg

WEEE / person / year

~ 43 smartphones



14.4 million tonnes

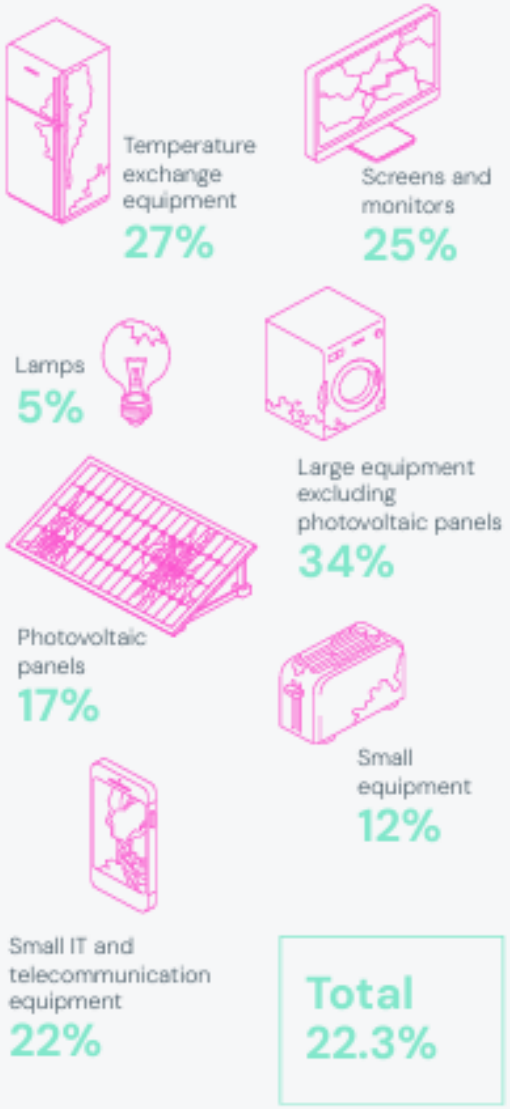
of electrical & electronic equipment is put on the market

5 million tonnes

of e-waste was collected in 2022

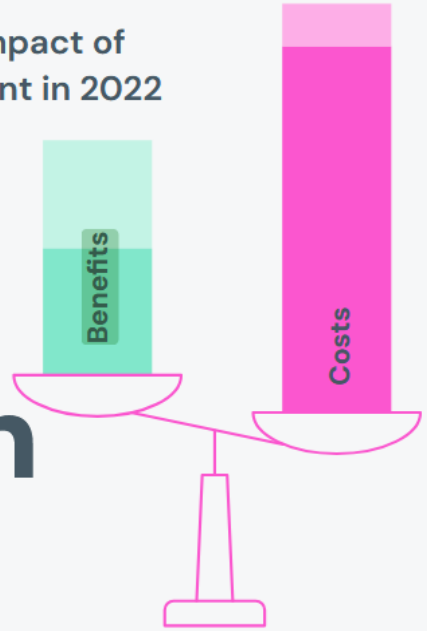
Sources : https://news.un.org/fr/story/2024/03/1144341, Parlement Européen

Figure 10. Documented Formal E-waste Collection and Recycling Rates by Category (2022)



Source: The Global E-waste Monitor 2024

Overall Economic Impact of E-waste Management in 2022



-37 billion USD

Annual economic monetary impact of e-waste management globally.

Benefits

23 billion USD
of monetized value of avoided greenhouse gas emissions.

28 billion USD
worth of recovered metals brought back into the circular economy.

Costs

10 billion USD
associated to the cost for treatment of e-waste.

78 billion USD
in externalized costs to the population and the environment.

Source: The Global E-waste Monitor 2024

Sources :The Global E-waste Monitor 2024



57 % de pertes

La majorité des matériaux recyclés sont perdus au cours du processus. Sur les 13,8 milliards de kg d'e-déchets collectés chaque année, nous perdons 7,8 milliards de kg de matériaux (dont 1 milliard de kg de métal).



Sources :Ifixit, Global E-waste

1/REPARE

2/REUSE

3/RECYCLE

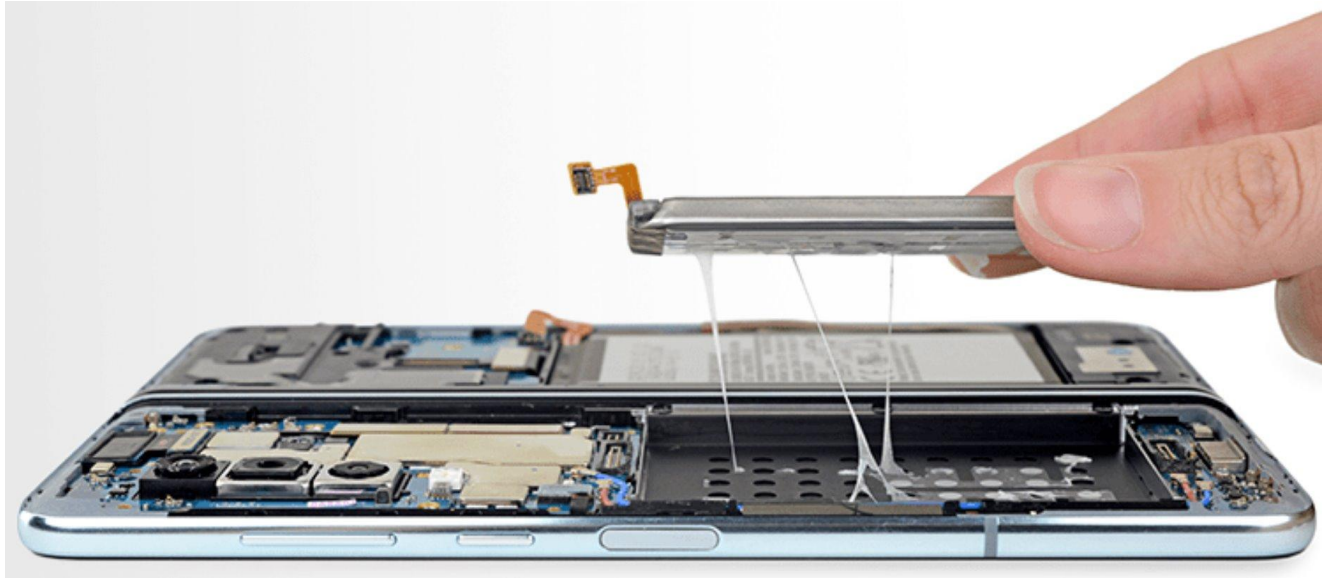
4/WASTE

1/ REPARE

To increase the lifetime of digital product and avoid waste.

Repairable products must be repaired.
Refurbished phones can be resold.
Repaired computers help the digital divide.

Better for the economy :
The repair professions are regional.
They will never be outsourced.



1/ REPARE : not so easy

Some restrictions within conception

- glue batteries together with strong adhesives.
- proprietary screw heads.
- software links parts to your device's serial number or logic board, causing bugs or blocked when you change a part

1/ REPARE : not so easy

For a company, they need to change the way it works :

- Train people
- Dedicate time, space
- Have spare pieces in stock

- Evaluate the impacts : contract, manufacturer support, capacity, economy/environment



What is your repair score ?

> Go to Wooclap <

2/ REUSE

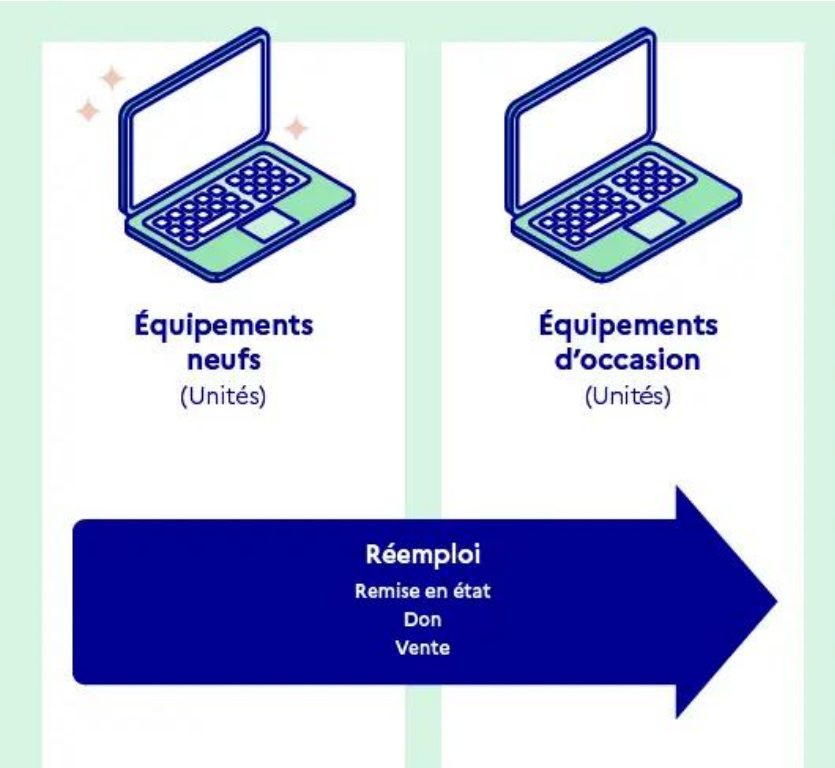
To give digital products a second life and actively act against waste.

For a company, they can reuse in different situations :

- Spare stock
- for interns or temporary workers
- for other function

They can also work with company or association to sell or give their functional equipments and reduce digital divide and digital waste.

2/ REUSE

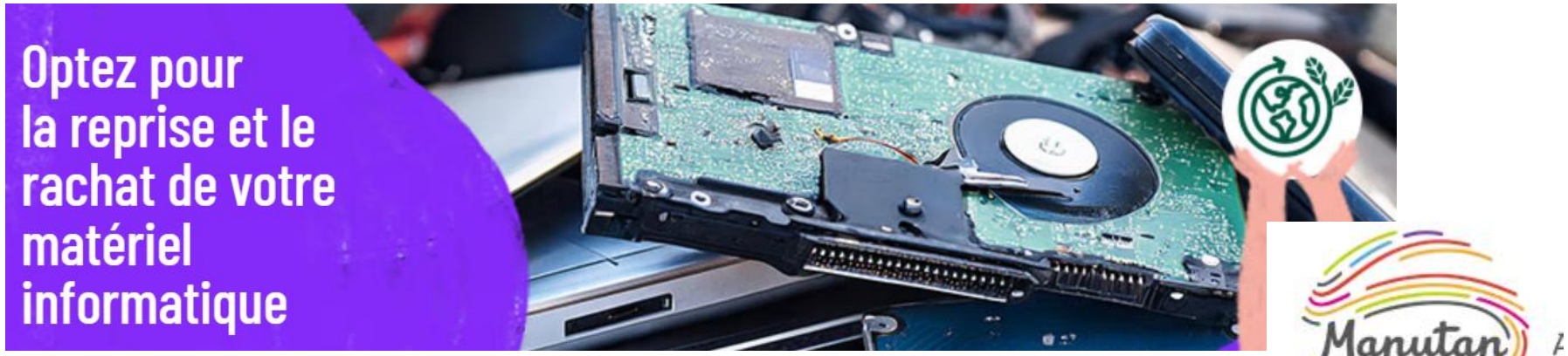


Main options to reuse outside your company :

- **Sell products**
- **Propose to your employees**
- **Give**

2/ REUSE

- **Sell your products to a broker**
- **Take over by your supplier**



https://www.manutan.fr/fr/maf/revalorisation-produits-electroniques

2/ REUSE

Offer a second life by selling products to your employee

⚠️ support the process to avoid over equipment and wrong sort at the end of life



RE Sources

https://re-sources.pro/

2/ REUSE

Giving your organization's digital equipment a second life through solidarity

♻️🌿 Supporting social and solidarity economy : refurbish are mainly social association

la collecte .tech avec EMMAÜS CONNECT

https://lacollecte.tech/



Leroy Merlin Example

La fin de vie du matériel informatique chez Leroy Merlin France

Visite de l'atelier ASF, partenaire d'Econocom, notre distributeur de matériel

ATELIERS SANS FRONTIERES

Ateliers Sans Frontières (ASF) est un **chantier d'insertion** créé en 2003 qui accueill
120 jeunes et adultes fragilisés par an, pour les aider à construire leur projet de vie, à
leur dignité et les amener à une situation personnelle et professionnelle stable.

ASF promeut l'insertion par des **activités solidaires à forte portée soc**
environnementale (recyclage, économie circulaire, don de matériel revalorisé) qui do
sens au travail effectué par les salariés et participent à leur motivation.

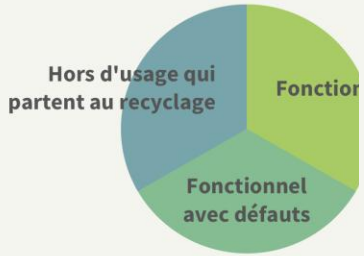
En quelques chiffres :

- **78 000** éléments informatiques traités en 2018
- **125** personnes accompagnées en 2018
- **70%** de sorties dynamiques

Activités réalisées :

- DEEE : collecte et recyclage de matériel informatique
- Logistique e-commerce pour des acteurs de commerce équitable

Chez Leroy Merlin :

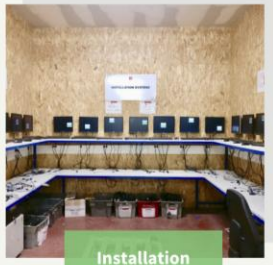


Plus de 6000 équipements DEEE collect
Econocom chaque année.

Vers une seconde vie du matériel informatique

(matériel encore fonctionnel)


Test matériel


Installation


Reconditionnement

Le recyclage du matériel informatique

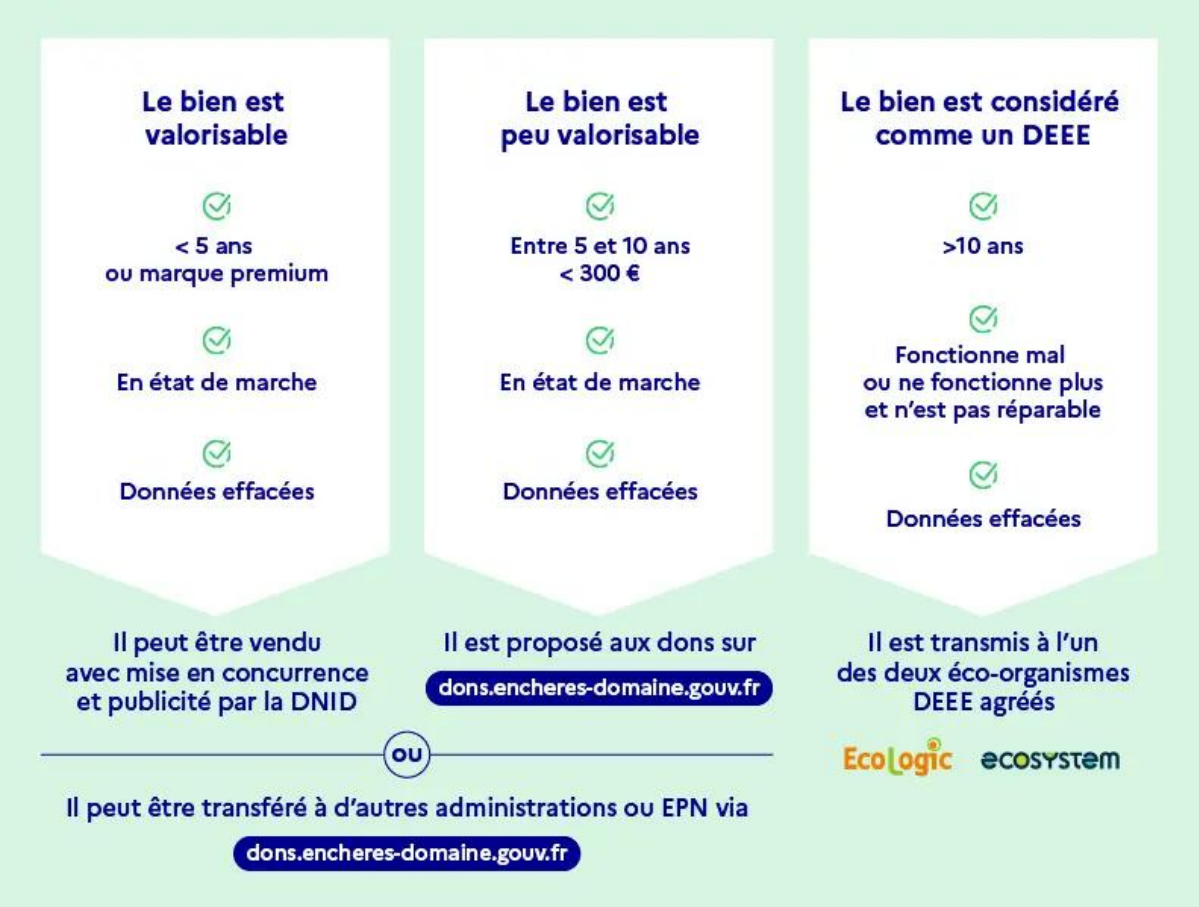
(matériel hors d'usage)


Tri à l'unité
Disque dur


Tri à l'unité
Carte mère, câbles...


Tri à l'unité
Ventilateur

Process for French public administrations



https://ecoresponsable.numerique.gouv.fr/publications/fin-usage/don/



REGULATION

Some laws to support reparation & reuse

France :

- Law AGEC
- Law REEN

Europe :

- Regulations (EU) 2023/1670 (ecodesign) and 2023/1669 (energy labelling)

https://www.indicereparabilite.fr/

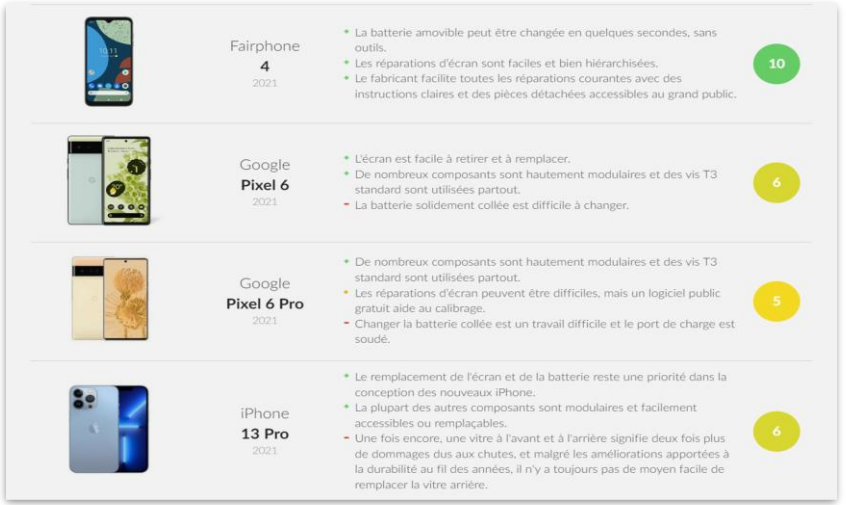


- **'AGEC' Law (Anti-Waste for a Circular Economy)**

Reparability Index (2022) and Durability Index (2024)
Spare parts for more than 5 years, Updates for 2 years
Public authorities must have 20% of their purchases reconditioned or made with recycled materials
Obligation to provide information on the period during which software updates provided at the time of purchase remain compatible with normal use of the device.
Repair bonus (End of 2022)

- **'REEN' Law (Reducing the Environmental Footprint of Digital Technology)**

Raising awareness of digital sobriety (primary-university)
Training in ecodesign and LCA of digital services (engineering qualification)
It is forbidden to restricte consumers' freedom to install the software or operating systems of their choice

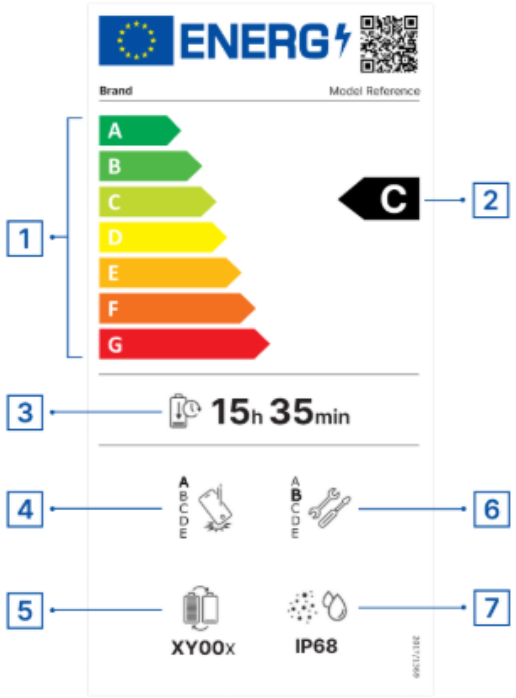


https://fr.ifixit.com/smartphone-repairability

Regulations (EU) 2023/1670 (ecodesign) and 2023/1669 (energy labelling), become effective on **20 June 2025** :

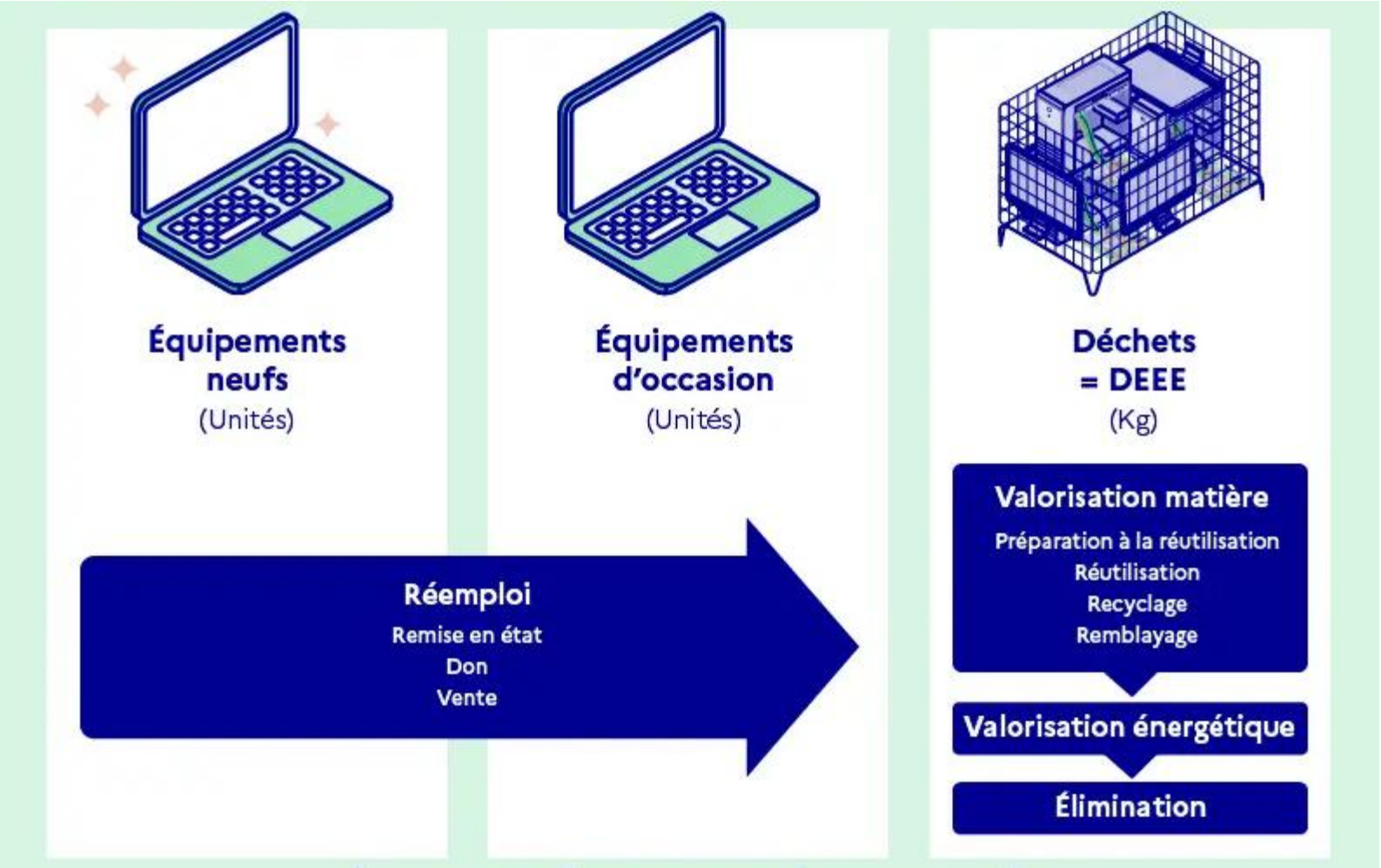
They cover the "ecodesign requirements applicable to smartphones, mobile phones, cordless phones and tablets" and the requirements for displaying related information.

Understanding the Smartphones and Tablets Energy Label



1. Scale of energy efficiency classes from A to G.
2. The energy efficiency class of this product.
3. The battery endurance per cycle, in hours and minutes per full battery charge.
4. Repeated free fall reliability class.
5. Battery endurance in cycles.
6. Repairability class.
7. Ingress Protection rating.

https://energy-efficient-products.ec.europa.eu/product-list/smartphones-and-tablets_en



https://ecoresponsable.numerique.gouv.fr/publications/fin-usage/cycle-de-vie/

3 / RECYCLE

The challenges of recycling :

- Preventing pollution
- Preserving resources
- Promoting local jobs
- Meeting the obligations of European Union member states

Collection rate : 49% (all WEEE)



3 / RECYCLE

There are different types of waste recovery, (precise definitions set out in the Environmental Code) :

REUSE *(réemploi) S*ubstances, materials, or products that are not waste are **used again for the same purpose** for which they were designed.

REUSE WASTE *(réutilisation)* Substances, materials, or products that have become **waste are used again.**

RECOVERY *(valorisation)* **Waste is used for a useful purpose as a substitute** for other substances, materials, or products.

MATERIAL RECOVERY: **any recovery operation other than energy recovery.** It includes preparation for reuse, recycling, backfilling, and other forms of material recovery, such as the **reprocessing of waste into secondary raw materials** for engineering purposes in road construction and other infrastructure projects.

RECYCLING: any recovery operation by which **waste is reprocessed into substances**, materials, or products for its original purpose or for other purposes.

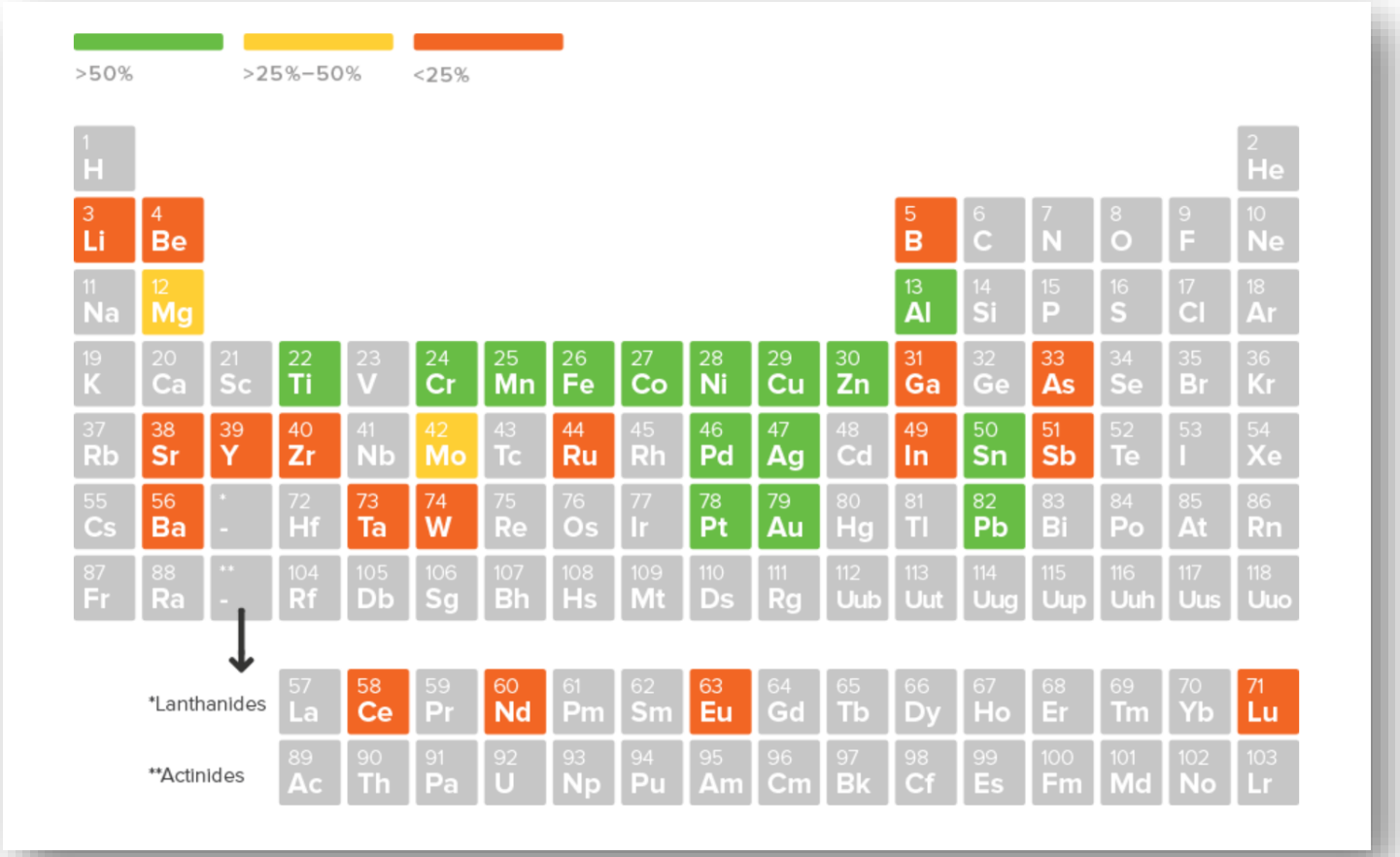
⚠Energy recovery from waste, waste-to-fuel conversion, and backfilling operations cannot be classified as recycling. ⚠️

3 / RECYCLE

“E-waste recycling does not meet more than 1% of the demand for essential rare earth elements. In other words, the status quo cannot continue”

Kees Baldé, lead author of UNITAR - United Nations Institute for Training and Research (UNITAR)

3/ RECYCLE : it is not always possible



50% of the 30 metals in your cell phone, have a functional recycling rate of less than 50%.

>> the metal's useful properties cannot be retained and it cannot be reused.

4/ WASTE

Professional WEEE (Waste Electrical and Electronic Equipment) is classified into several categories, ranging from office equipment to household appliances.

This classification, defined by the professional WEEE regulations, helps organize the separate collection, recycling, and management of WEEE generated by professional activities.

Adherence to the WEEE waste code is crucial to ensuring the proper and environmentally sound treatment of each WEEE waste category.

WEEE : Waste of Electric and Electronic Equipments

Category		Examples
1	Temperature exchange equipment (TEE)	Refrigerators, Air-conditionning
2	Screens, monitors, and equipment containing screens having a surface greater than 100 cm2	Monitors, Laptops
3	Lamps	
4	Large equipment (any external dimension more than 50 cm)	Washing machines
5	Small equipment (no external dimension more than 50 cm)	Microwaves, electric shavers
6	Small IT and telecommunication equipment (no external dimension more than 50 cm)	Mobile phones, GPS, routers...
7	Photovoltaic panel	

4/ WASTE

When does electrical or electronic equipment (EEE) become waste?

Electrical or electronic equipment becomes waste electrical and electronic equipment (WEEE) for the following reasons:

- it is no longer functional;
- it is repairable but the cost of repair is prohibitive;
- it is part of a larger system, one component of which is no longer functional;
- it is functional but obsolete and has been replaced by newer equipment.

4/ WASTE : WHAT IS AN ECO-ORGANIZATION

Obligations
- Non-profit organization
- Commitment to qualitative and quantitative objectives
- Obligation of transparency
- Continuous improvement process
- Transfer of responsibility

Commitments
- Commitment to achieving results on behalf of producers
- Guaranteeing a healthy supply chain

Characteristics
- Meets strict specifications
- Submits its application to all stakeholders in the supply chain
- Is accredited by government ministries
- Is audited during the accreditation process by all stakeholders

4/ WASTE : ECO-ORGANIZATION FOR DEEE

Waste management systems, guaranteeing extended producer responsibility (EPR)

This system allows for the integration of a part of the environmental impact of consumer products into their price. EPR places the financial costs of the waste disposal during the purchase on the consumer.

EcoLogic

ecosystem
recycler c'est protéger



ECOLOGIC WASTE COLLECTION FROM 2007 IN FRANCE



Source : Ecologic

Waste management :

TO BETTER PROTECT PUBLIC HEALTH AND THE ENVIRONMENT

Extracting hazardous or regulated substances

One of the key reasons for the existence of our sector is to protect public health. ecosystem's objective is to extract and then neutralize the hazardous substances contained in the appliances collected, an activity which has required the creation and development of specialised treatment sites in France. In 2023, ecosystem collected 735 343 tonnes of Lamps, Household and Professional WEEE in France. 62 521 tonnes of hazardous or regulated substances were extracted.

QUANTITIES OF HAZARDOUS SUBSTANCES EXTRACTED IN 2023

201 tonnes
Of toner cartridges

442 kg
Of components containing mercury

413 tonnes
Of capacitors containing polychlorinated biphenyl (PCB) and electrolytic capacitors containing hazardous substances

19 392 tonnes
Of LCD screens and screens backlit by discharge lamps

15 984 tonnes
Of plastic materials containing bromide-based flame retardants

1 684 tonnes
Of batteries and accumulators

7 708 tonnes
Of cathode ray tubes





focus

DECONTAMINATING LCD SCREENS

Plasma panels contain elements such as calcium, sodium, barium and strontium which can cause skin irritations and potentially damage the eyes.
These elements can also be corrosive.
In addition, like older generation LCD screens, plasma screens are backlit using lamps containing mercury. Mercury can be extremely harmful to the body, namely at the level of the nervous system and kidneys. It also interferes with reproduction and development.

Source : Ecosystem

BEST PRACTICES SUMMARY

BEST PRACTICES

- Last your digital equipment life : take care, repare...
- Reuse by refurbishing
- Reuse by donating working equipment
- Reuse by selling working equipment
- Use an eco-organization for WEEE management

INCLUSIVE DIGITAL



Few figures about inclusive digital

More than **18% of the French population has a disability** (APF France Handicap, 2023).

In 2024, **less than 10% of websites in France were fully accessible** to people with disabilities (Ministry of Solidarity and Health).

60% of French respondents believe they have lost customers due to a lack of accessibility, and **38% believe their company's reputation could suffer** if this issue is not addressed within the next 12 months (Capterra, 2024).

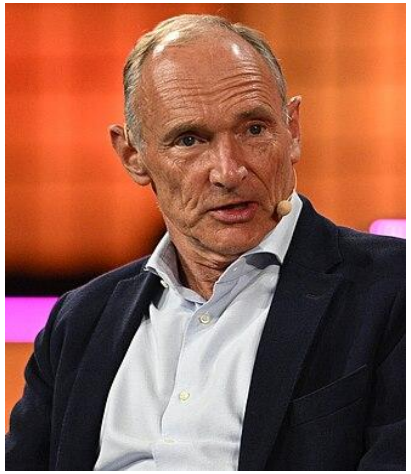
In Europe, **public sector websites score 44% highe**r than those in the private sector (Contentsquare, 2024).

The e-commerce and media sectors lag behind in terms of accessibility, often meeting less than 50% of accessibility criteria (Contentsquare, 2024).

https://www.fevad.com/les-chiffres-sur-laccessibilite-web/

"The power of the Web lies in its universality. Access for all, regardless of disability, is an essential aspect.“

Tim Berners-Lee, director of the W3C and inventor of the World Wide Web



"Le pouvoir du Web est dans son universalité.
L’accès pour tous, quel que soit le handicap, est un aspect essentiel.“

Tim Berners-Lee, directeur du W3C et inventeur du World Wide Web

Accessibility, a fundamental right

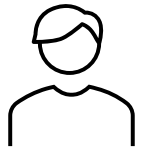
"One of the fundamental objectives of the W3C is to make the Web and its services available to all individuals, regardless of their hardware or software, network infrastructure, native language, culture, geographic location, or physical or mental abilities."

The W3C (World Wide Web Consortium) was created on October 1, 1994.

“Persons with disabilities are defined as those who have long-term physical, mental, intellectual or sensory impairments which, in interaction with various barriers, may hinder their full and effective participation in society on an equal basis with others.”

Article 1 – UN Convention on the Rights of Persons with Disabilities – 13 December 2006

Access internet, few steps



Who ? Capacity ? From where ?



Bandwith ? Costs ? Connectivity ?



Which terminal ? Which OS version?



Which service ?

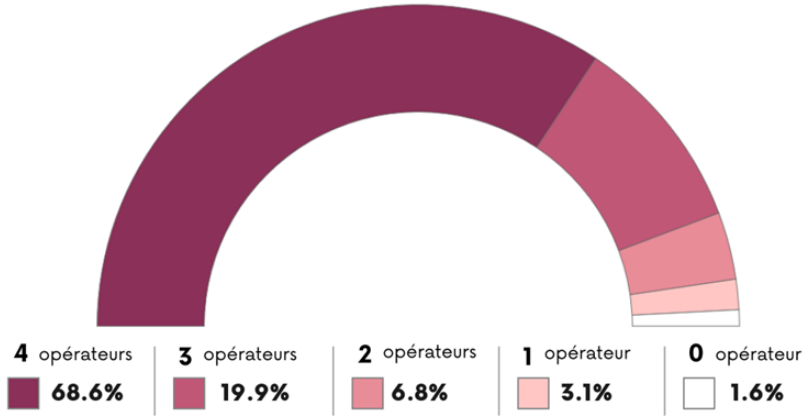
Technical Accessibility

Population couverte en **internet mobile** en intérieur

88.5% de la population est couverte **par au moins 3 opérateurs.**

Population couverte par opérateur	Bouygues	Free	Orange	SFR
	93.9%	77.6%	92.8%	86.5%

Population couverte selon le nombre d'opérateurs



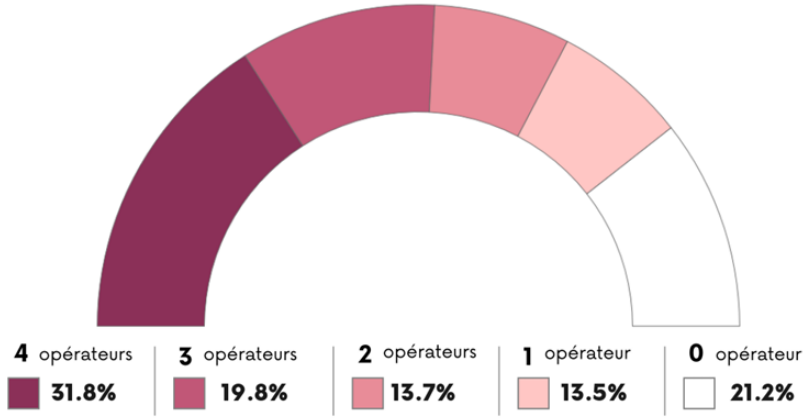
Sources : Arcep, 2025.

Superficie couverte en **internet mobile** en intérieur

51.6% de la superficie est couverte **par au moins 3 opérateurs.**

Superficie couverte par opérateur	Bouygues	Free	Orange	SFR
	65.9%	43.4%	61.2%	57.2%

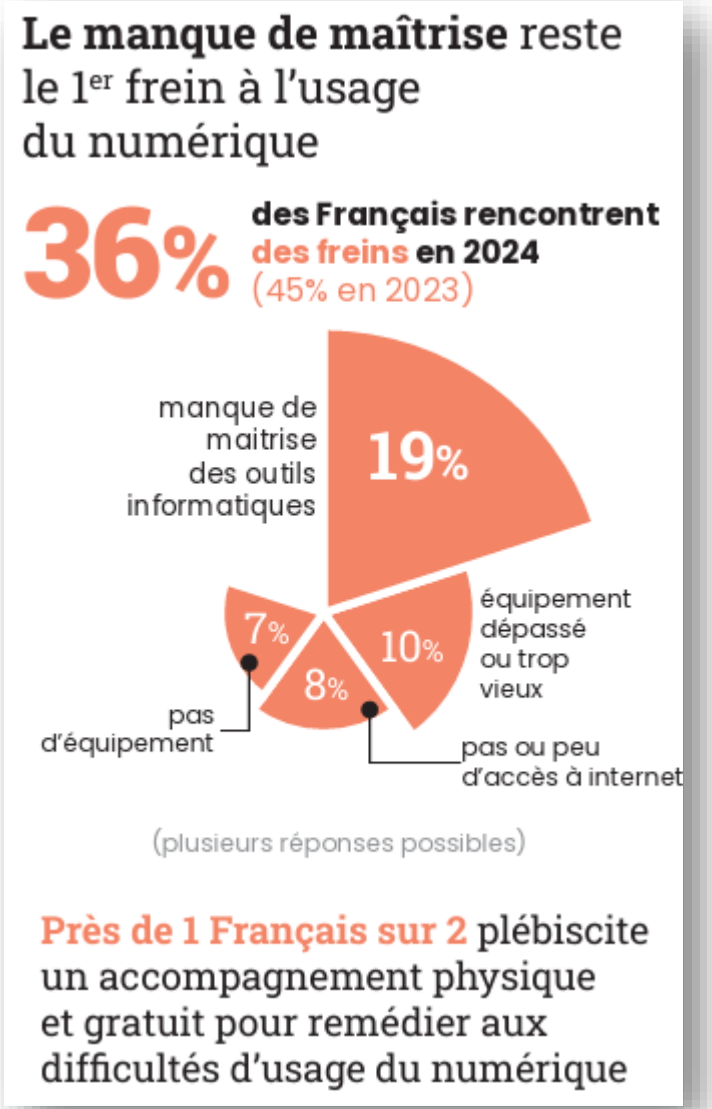
Superficie couverte selon le nombre d'opérateurs



Sources : Arcep, 2025.

https://www.arcep.fr/cartes-et-donnees/nos-cartes/la-couverture-mobile-par-departement.html

Technical skills



https://anct.gouv.fr/ressources/publication-barometre-du-numerique-2025

1 in 5 people has limitations in their activities

Different Types of Disabilities

- Physical and Sensory : visual, auditory, and motor impairments.
- Mental intellectual, cognitive (e.g., learning disabilities), and mental health disabilities.
- Multiple Disabilities: a motor or sensory disability combined with a mental disability.
- Disabling Health Conditions: cancer, chronic pain, allergies, etc.

Everyone is different.

Digital technology as a vector of inclusion

- **Autonomy**

No longer depend on someone else's help for simple tasks.

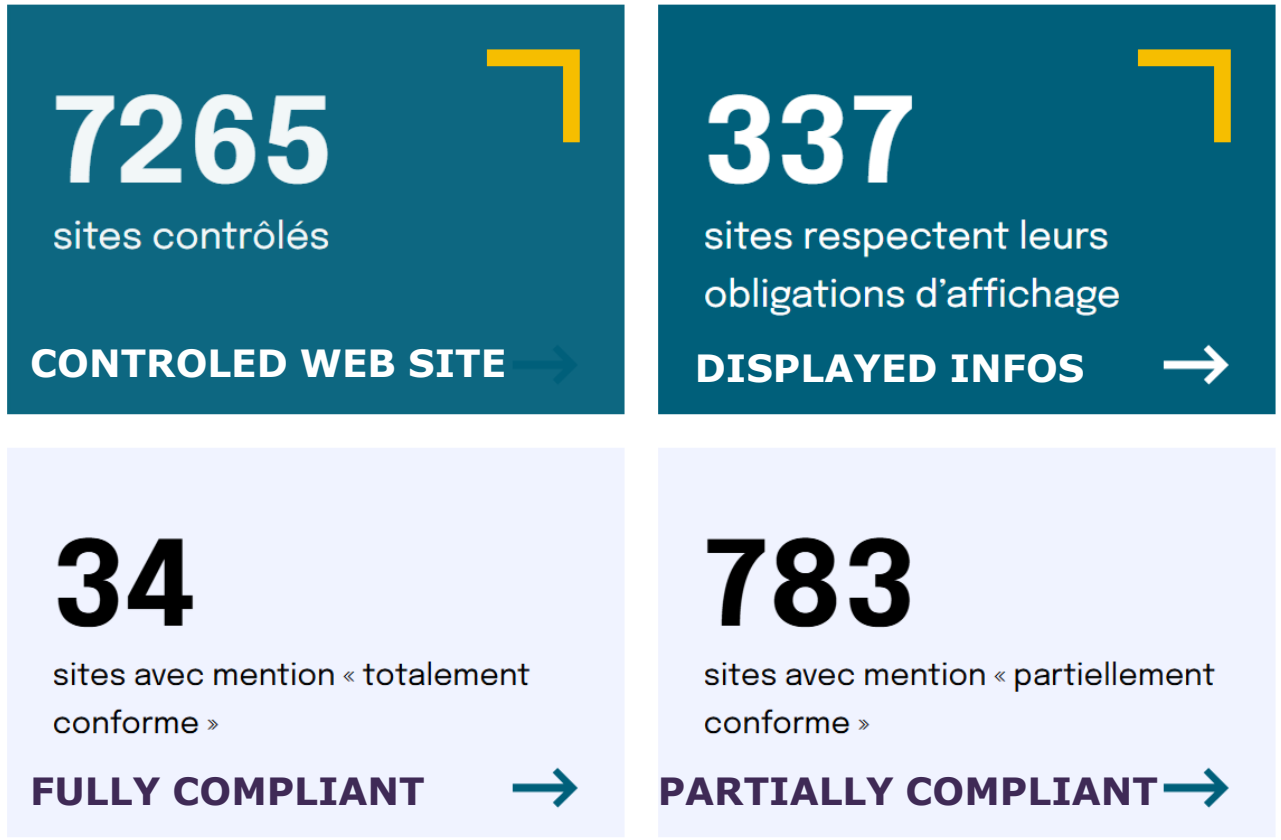
- **Integration**

Facilitate professional life and improve access to employment for people with disabilities.

- **Equality**

Use the same means of expression as everyone else (social networks, etc.).

Digital technology as a vector of inclusion



https://ants.gouv.fr/accessibilite-des-demarches-en-ligne-et-des-applications-mobiles
https://observatoire-access-num.aveuglesdefrance.org/

Specific needs for internet browsing

- Browse using only the keyboard
- Browse using only the mouse
- Read descriptions of visual elements
- Read descriptions of audio or video elements
- Browse at your own pace
- Customize the display (colors, font size, font, etc.)





Use these simulators
to test the experience of all users…

https://www.atalan.fr/agissons/fr/index.html

https://simulation-accessibilite.inria.fr/index.html

Accessibility standards - W3C

- UAAG - User Agent Accessibility Guidelines
for user agents, that is, the software used to access the web.

- ATAG - Authoring Tool Accessibility Guidelines
For content editing tools

- WCAG - Web Content Accessibility Guidelines
For web content

- ARIA - Accessible Rich Internet Applications
For rich components



Focus on the WCAG standards

It is a technical standard
Three accessibility levels : **A**, **AA**, and **AAA**.

The levels are determined based on two criteria:

- importance for people with disabilities
- feasibility.

Level A ensures access to information and basic usability.

Level AA includes improvments and support for specific contexts.

WCAG 2.2 standard : 4 principles

- **Perceivable** - Information and user interface components must be presentable to users in ways they can perceive.
- **Operable** - User interface components and navigation must be operable.
- **Understandable -** Information and the operation of user interface must be understandable.
- **Robust -** Content must be robust enough that it can be interpreted reliably by a wide variety of user agents, including assistive technologies.



If any of these are not true, users with disabilities will not be able to use the web.

https://www.w3.org/TR/WCAG22/#toc

ACCESSIBILITY



REGULATION

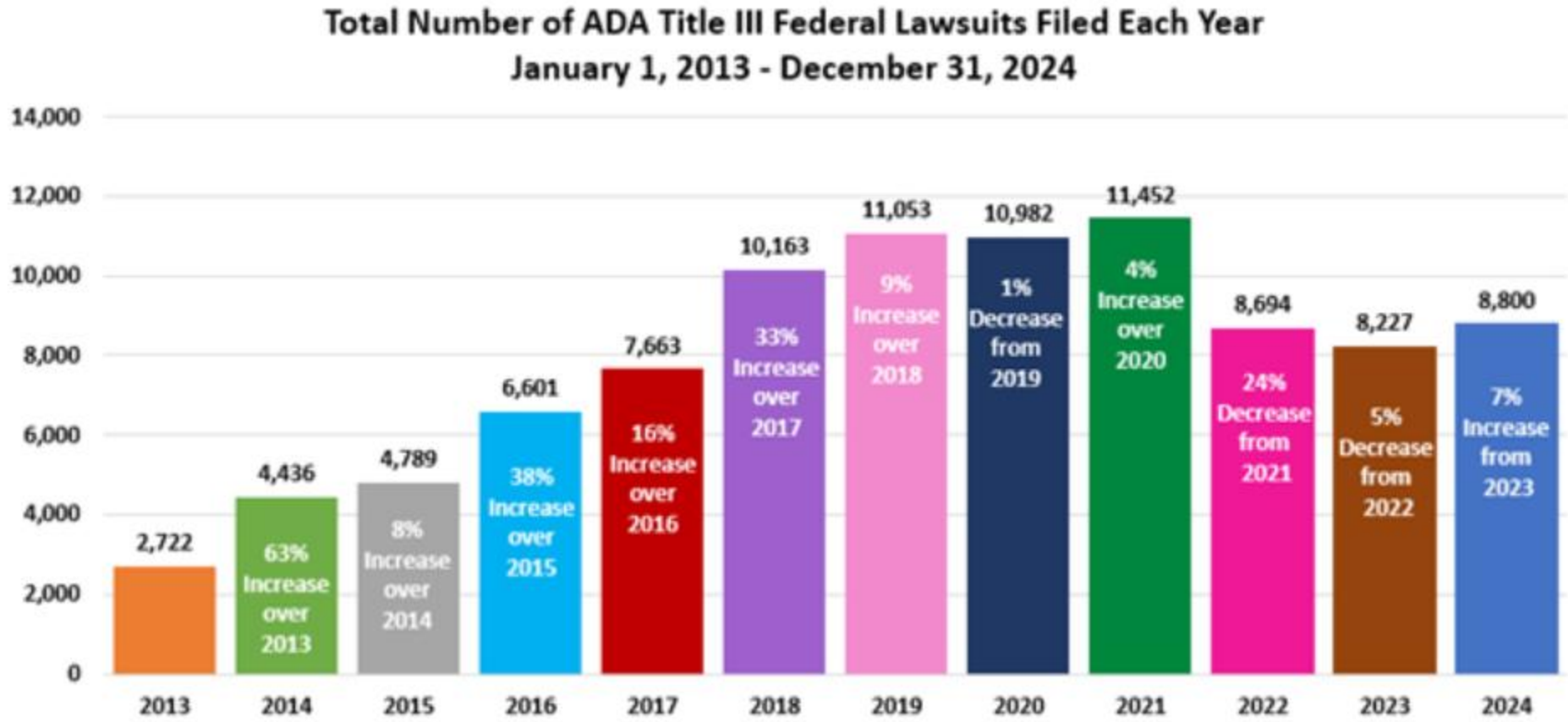
US : Americans with Disabilities Act (ADA)
https://www.ada.gov/

Europe : European Accessibility Act (EAA)
https://commission.europa.eu/strategy-and-policy/policies/justice-and-fundamental-rights/disability/union-equality-strategy-rights-persons-disabilities-2021-2030/european-accessibility-act_en

France :

- article 47 de la loi du 11 février 2005 sur l'égalité des chances
- 2009 première version RGAA
- décret n° 2019-768 du 24 juillet 2019

In 2024, 8 800 ADA Title III complaints have been filled in federal district courts.



https://www.adatitleiii.com/2025/03/ada-title-iii-federal-lawsuit-numbers-rebound-to-8800-in-2024/

European Accessibility Act - EAA



The European Accessibility Act (EAA) became effective on June 28, 2025.

This directive aims to harmonize accessibility rules across all European Union countries, imposing obligations on the private sector in the same way as the public sector.

- Any breach of the obligations arising from this Act will be punishable by a fine of €7,500, which can increase to €15,000 in the event of a repeat offense.
- Private companies with fewer than 10 employees and an annual turnover or balance sheet total not exceeding €2 million are exempt from these new obligations.
- Approximately 87 million Europeans with disabilities are affected by the European Accessibility Act.

French key dates for Accessibility



February 11, 2005
Digital accessibility becomes a legal requirement for public sector websites.

From 2012
All government and local authority websites must comply with the RGAA (General Accessibility Guidelines for Administrations).

October 7, 2016
Extension of the legal requirement to a portion of the private sector, subject to a revenue threshold (250 million euros).

July 24, 2019
Transposition of the European regulatory framework and introduction of exemptions.

March 9, 2023
Transposition of European Directive (EU) 2019/882 on the accessibility of products and services, applicable from June 28, 2025.

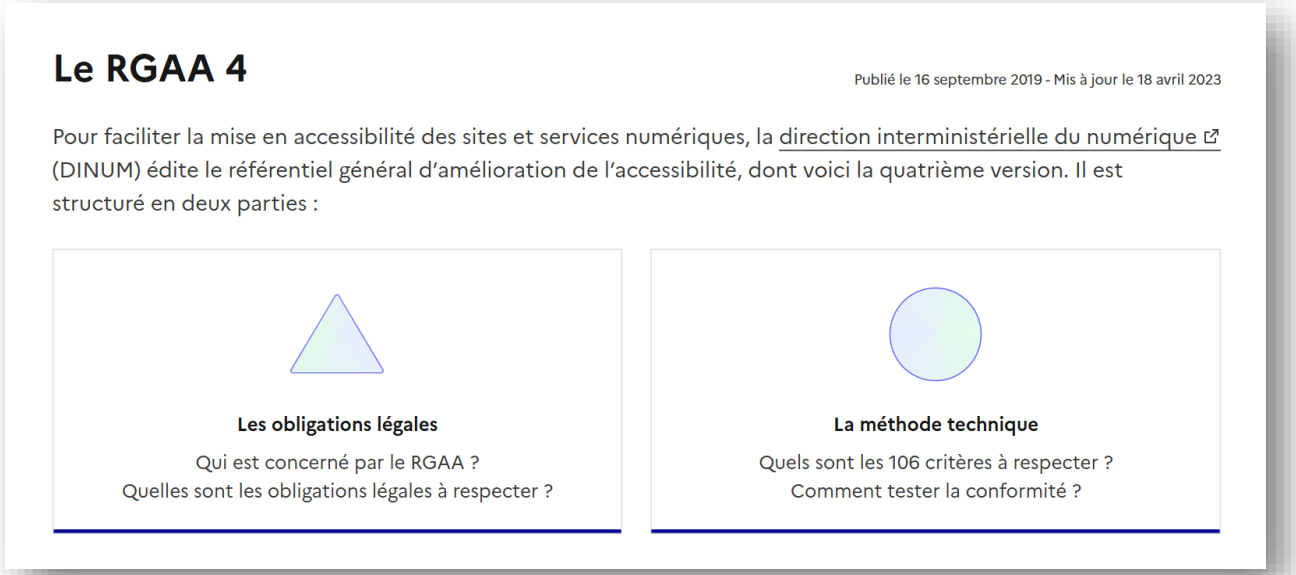
September 6, 2023
Stricter penalties for the public sector and establishment of a supervisory authority. Arcom.

October 9, 2023
Extension of the legal obligation to certain sectors (banking, transportation, telecommunications, audiovisual, e-books, and e-commerce) and establishment of five additional regulatory authorities: DGCCRF, ARCEP, ACPR, AMF, and Banque de France.

Conformité ≠ Accessible

Focus on RGAA, version 4.1.2

Référentiel Général d'Amélioration de l'Accessibilité





https://accessibilite.numerique.gouv.fr/methode/criteres-et-tests/

Top accessibility web issues

1. Ignoring alt tags for images
You see an image? A screen reader doesn't "see" it. It has activated the alt tag. Without it, a blind person would miss the message.

2. Using insufficient color contrast
Light gray on white? These WCAG contrast errors make the text unreadable for many. This includes those who simply have a less-than-ideal screen.

3. Forgetting keyboard navigation
Some people navigate using the Tab key. If your site skips elements or freezes… it's a nightmare.

4. Using out-of-order headings
Screen readers follow headings to structure the information. If you confuse them, they get lost. The hierarchy of headings is the table of contents of a website.

5. **Accessibility of interactive elements: forms, links, and buttons**
Form fields, links, and buttons are key interactive elements of a website. Each of them requires labels or descriptions to be accessible to all users.

Some tools for web quality and digital accessibility

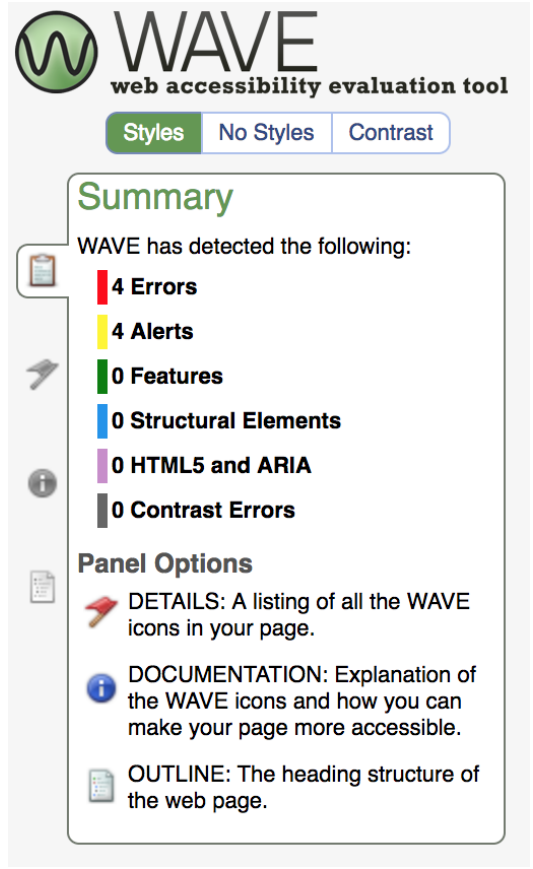
Sites

Check-lists Opquast : https://checklists.opquast.com/fr/assurance-qualite-web/
Tanaguru contrast finder : https://contrast-finder.tanaguru.com/
W3C Validator : https://validator.w3.org/

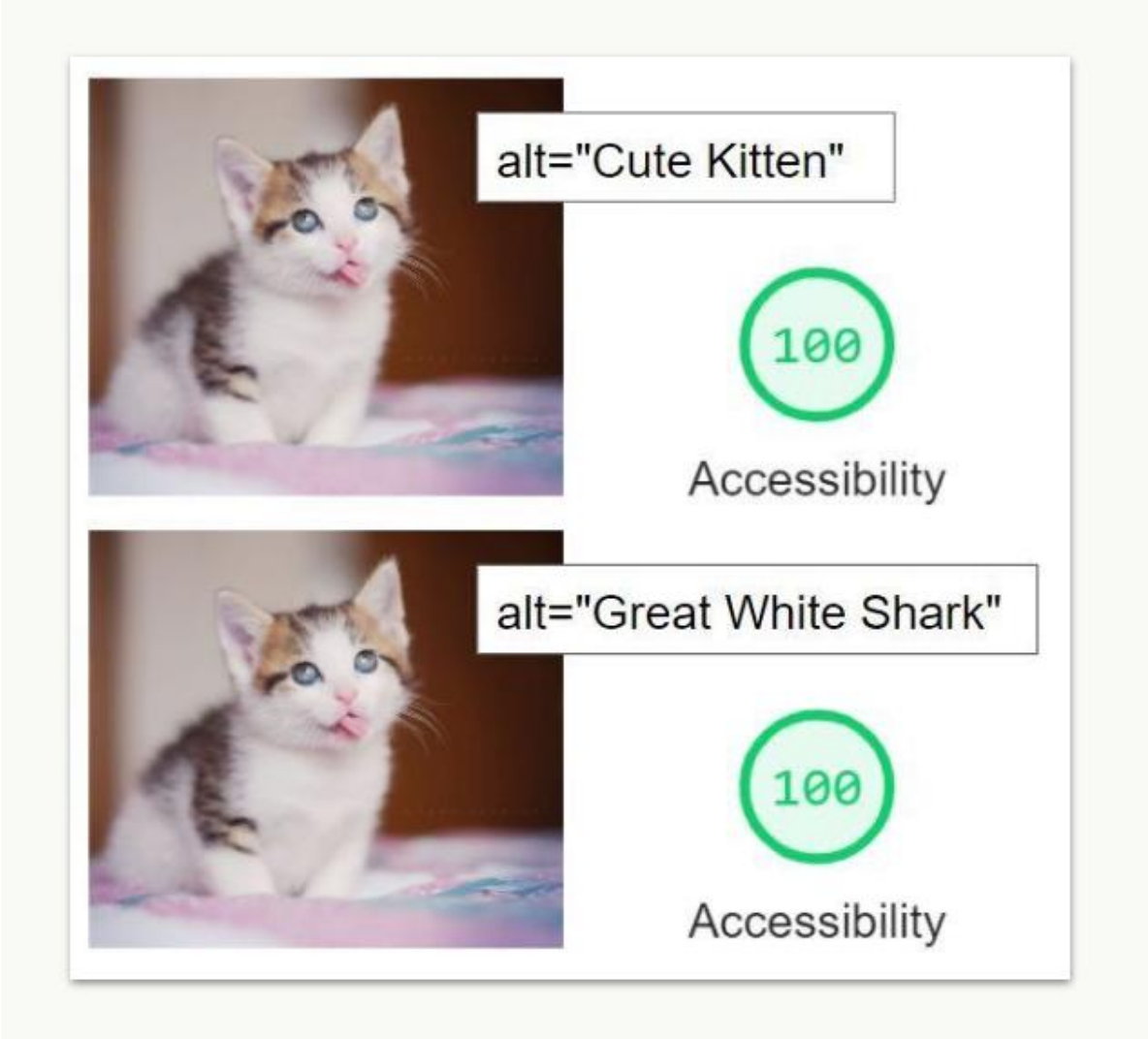
Plugins

WAVE
toolbar gives you a very quick first impression about the quality of the current webpage's accessibility. https://wave.webaim.org/

Lighthouse
Chrome puglin which offers performance and accessibility audits - https://pagespeed.web.dev/?hl=fr



Beware of automating accessibility tests.



Some examples

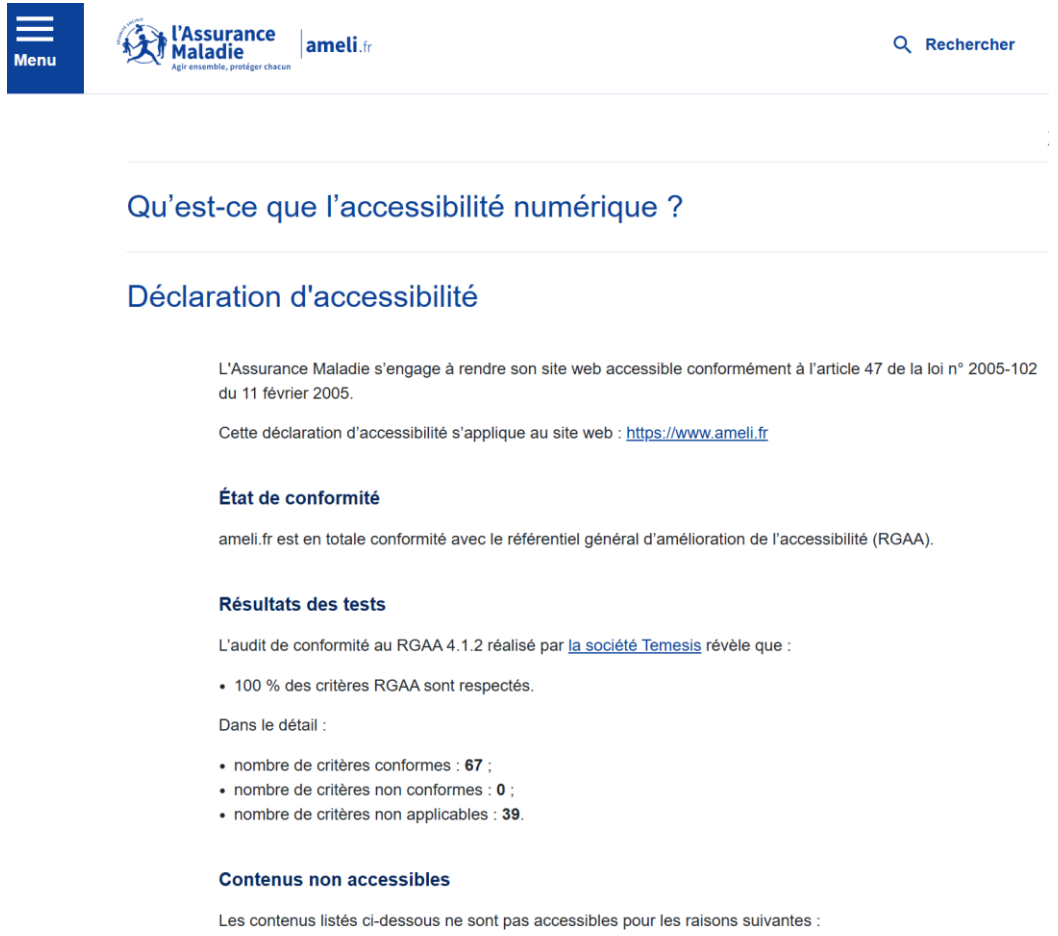
https://www.numerique.gouv.fr/accessibilite/

https://totalenergies.com/fr/accessibilite

https://www.laredoute.fr/accessibilite.aspx

https://www.fnac.com/declaration-accessibilite-web

https://www.ameli.fr/assure/accessibilite





By group 2 or 3, choose a website

Look about accessibility and make a quick check.

What are your observations ?

"The power of the Web lies in its universality. Access for all, regardless of disability, is an essential aspect."

Tim Berners-Lee, directeur du W3C et inventeur du World Wide Web

Lesson 5 - As a summary

- **End of life**
  - Materiality of digital
  - REPARE – REUSE
  - RECYCLE – WASTE
  - Respect regulation about WEEE

- **Accessibility**
  - Web is for everyone - W3C
  - Respect regulation for web accessibility : ADA - EAA



Think about impacts of your action
Think about disabilites during conception

JUNIA Grande école d'ingénieurs

HEI·ISEN·ISA

Septembre -Décembre 2025

ISEN
Module
Numérique Responsable